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Ruthenium-catalyzed dimerization of CF_3-containing functional allenes ☆

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ABSTRACT

An efficient method for the regioselective preparation of 1,3-dimethylenecyclobutanes based on Ru-catalyzed dimerization of readily available CF_3-containing functionalized allenes including α-amino acid derivatives have been developed. The reactions smoothly proceed in toluene at 100 °C in the presence of catalytic amounts (5 mol%) of naphthalene-containing ruthenium system $[Cp^*Ru(C_{10}H_8)]PF_6/[Et_3NBn]Cl$ to afford exclusively tail-to-tail coupling products in good to excellent yields.

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1. Introduction

Cyclobutanes are important structural motifs of many biologically active molecules including natural products [1] and are also valuable intermediates for a range of useful chemical transformations [2]. One of the straightforward and atom-economic access to cyclobutane ring construction is based on [2+2] cycloaddition of allenes [3]. Despite the thermal version of this transformation is well studied process [4], the control of regioselectivity (head-to-tail, tail-to-tail, head-to-head) [5] is often hard to achieve. On the other hand, less investigated dimerization of allenes catalyzed by transition metals (Ni [6], Pd [7], Au [8], Rh [9], Ru [10]) has been shown to be an efficient alternative strategy to provide the cyclobutane ring systems in regioselective manner. However, these catalytic methods mainly lead to 1,2-methylencyclobutanes with the only exception for Ru(II)-catalysis. Thus, Dixneuf and coworkers successfully realized tail-to-tail cycloaddition of allenyl boronate under catalysis with $Cp^*Ru(cod)Cl$ to establish the first direct pathway to regioselective formation of 1,3-methylencyclobutanes [10]. The interest to latter compounds deals with their great potential to be utilized as the key precursors for the generation of the corresponding cyclobutane-2,4-diradicals with triplet ground state to apply in material science as paramagnetic building blocks [11].

On the other hand, organofluorine compounds find nowadays more and more important applications in modern life and material sciences [12]. This fact stimulate the researchers to develop new efficient methodologies for the introduction of fluorine and fluorinated groups into different organic molecules.

In a recent study, we have developed an efficient one-step access to functionalized allenes **2** *via* [2,3]-sigmatropic rearrangement of propargyl-containing nitrogen ylides generated *in situ* from CF_3-containing diazo compounds **1** (Scheme 1A) [13]. The synthetic potential of **2** has been demonstrated by their inter- and intramolecular transformations under catalysis with transition metals [14], *e.g.* such as Cu-catalyzed hydroamination [14a] or Co-catalyzed Pauson-Khand reaction [13] to afford the corresponding α-amino acid derivatives (Scheme 1B). Now we wish to report on a convenient pathway to new trifluoromethyl-containing 1,3-dimetylencyclobutanes *via* dimerization of functionalized allenes under Ru-catalysis (Scheme 1C).

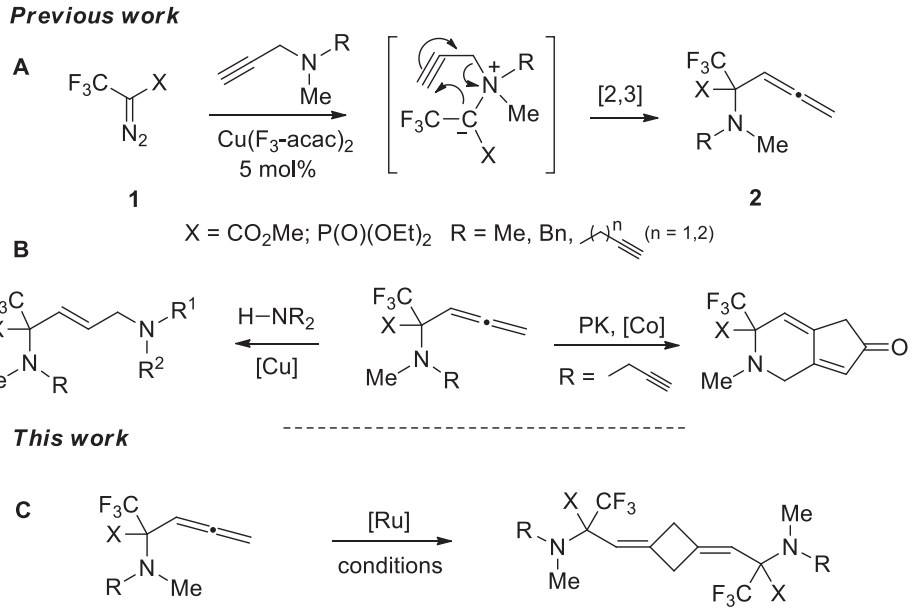
2. Results and discussion

Our continuous interest in the development of efficient methods for the preparation of diverse CF_3-containing compounds based on metal-catalyzed transformations of unsaturated precursors including alkynyl- and alkenyl α-amino acid derivatives

☆ This work is dedicated to Prof. Elena Shubina in recognition of her great contribution to organometallic chemistry.

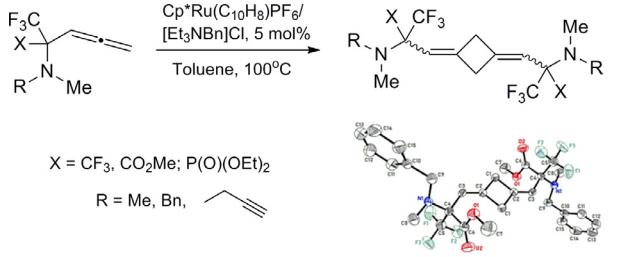
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Scheme 1. Previous and present work.

Table 1
Optimization of allene dimerization.[a].



Entry	Catalyst (mol%)	Solvent	T (°C)	Yield[b] (%)
1	Cp*Ru(cod)Cl (3)	DCE	85	78
2	Cp*Ru(cod)Cl (3)	toluene	90	82
3	Cp*Ru(cod)Cl (5)	toluene	95	98
4	[Cp*Ru($C_{10}H_8$)]PF_6/[Et_3NBn]Cl (3)	DCE	85	61
5	[Cp*Ru($C_{10}H_8$)]PF_6/[Et_3NBn]Cl (5)	toluene	90	80
6	[Cp*Ru($C_{10}H_8$)]PF_6/[Et_3NBn]Cl (5)	toluene	100	100/71[c]
7	-	toluene	110	0

[a] Reagents and conditions: allene **2a** (0.2 mmol), solvent (3 ml);
[b] Determined by ^{19}F NMR spectroscopy.
[c] Isolated yield.

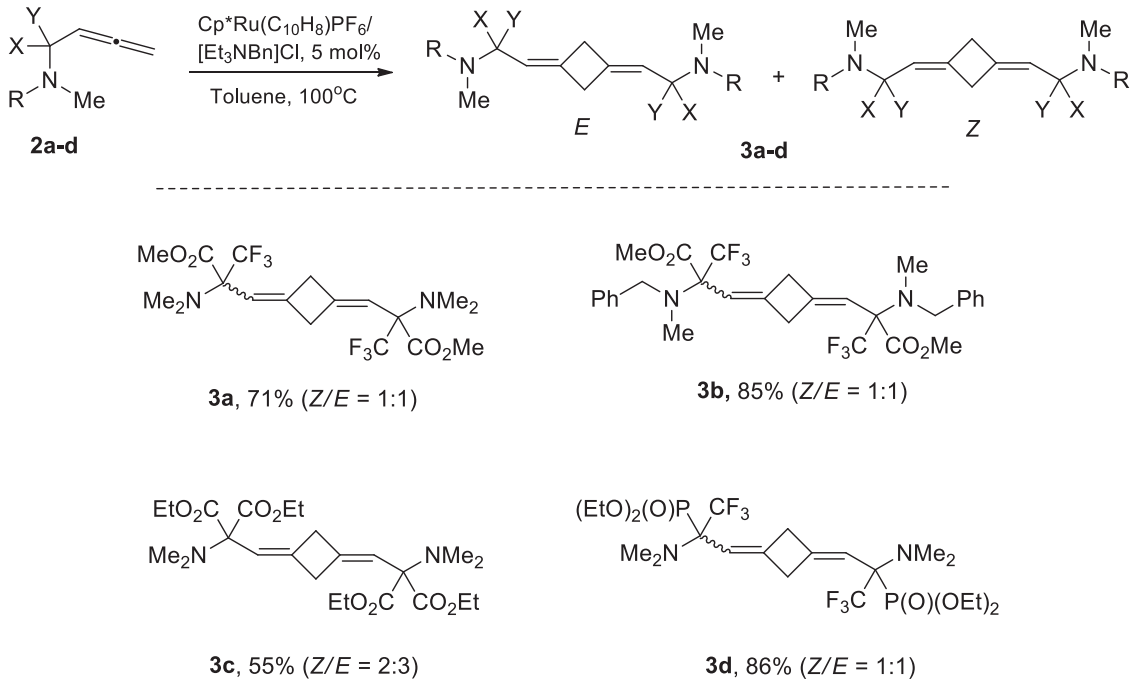
[15] prompted us to explore the reactivity of functionally substituted allenes in dimerization process catalyzed by Ru-complexes. In our initial attempt, being inspired by Dixneuf's finding [10], we examined the catalytic activity of Cp*Ru(cod)Cl in intramolecular [2+2] cyclization of allene **2a** chosen as model substrate. Thus, we found that the reaction takes place when boiling in dichloroethane (DCE) in the presence of 3 mol% of catalyst leading to formation of the desired cyclobutane **3a** in 78% yield for 2 hours (Table 1, entry 1). The replacing the solvent with toluene, increasing the temperature (up to100 °C) and catalyst loading up to 5 mol% led to 98% yield of **3a** (entry 3). Then we tested the activity of naphthalene complex [Cp*Ru($C_{10}H_8$)]PF_6, which generates the same active species [Cp*RuCl] in the presence of chloride ion [16]. This catalyst is a more convenient one to work with than [Cp*Ru(cod)Cl], because it is air-stable and easily accessible by using a one-step procedure from $RuCl_3 \bullet 3H_2O$ [17]. After variation of solvent, catalyst loading and reaction temperature the quantitative yield of **3a** has been achieved when the reaction was performed with 5 mol% of naphthalene-containing catalytic system [Cp*Ru($C_{10}H_8$)]PF_6/[Et_3NBn]Cl in toluene at 100 °C for 2 hours (entry 6). When the reaction was carried out in the absence of any catalyst, no product was observed (entry 7). Monitoring of the reaction in all cases was accomplished using ^{19}F NMR spectroscopy and TLC analysis.

As result, ruthenium catalytic system [Cp*Ru($C_{10}H_8$)]PF_6/[Et_3NBn]Cl was chosen as preferential catalyst for dimerization process. With the optimized conditions in hand, we examined the reactivity of three more allenes bearing N(Me)Bn (**2b**), ethyl carboxylate (**2c**) and phosphonate (**2d**) groups, which were prepared by the same way using previously described protocol (Scheme 1A). In all cases the corresponding 1,3methylenecyclobutanes **3b-d** were obtained in moderate to good yields (Scheme 2).

1,3-Methylenecyclobutanes **3a-d** were initially isolated as *cis-*, *trans*-isomer mixtures in a ratio indicated on Scheme 2 (determined by ^{1}H and ^{19}F NMR spectroscopy). The compounds **3a,b** and **3d** bearing asymmetric center additionally formed diastereomers in equal amounts approximately, which can be easily separated by column chromatography on silica gel. In the case of **3b** and **3d** we were successful to separate in addition *cis-* and *trans*-isomers in the individual forms (see Supporting Info). The characterization of the compounds obtained was performed using standard physicochemical methods and elemental analysis. The assignment of signals of *cis-* and *trans*-isomers in the mixtures was achieved by 2D NOESY NMR experiments. While for both isomers dipolar cross-peaks appears between olefine and cyclobutane protons, they are significantly more intense in the case of the *trans*-isomer, reflecting larger number of interacting nuclei (Fig. 1).

In addition, the structures of individual stereoisomers of **3a-c** were determined by X-ray diffraction (Fig. 2).

In the case of allenynes **2e** [13], **2f** [14c] and **2g** [14c], likely due to the presence of an additional triple bond capable to coordinate with catalytic species, the reaction leads to the complicated mixture of the products that were impossible to identify. Therefore, we decided to block triple bond by Cu(I)-catalyzed "click" re-



Scheme 2. Synthesis 1,3-methylencyclobutanes 3a-d.

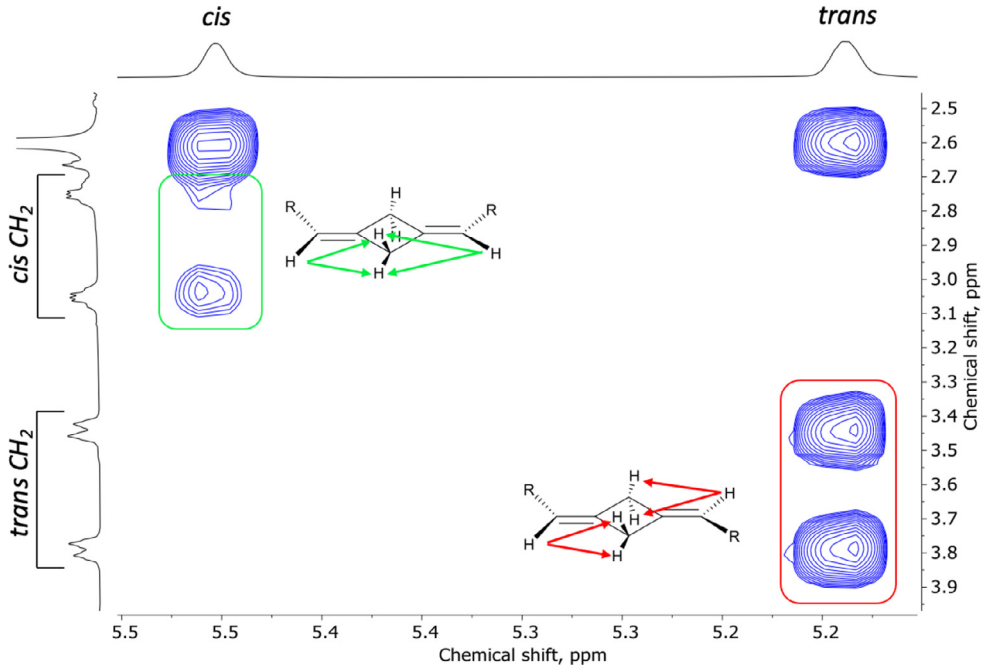


Fig. 1. A fragment of 2D ^{1}H NOESY NMR spectrum (600 MHz, $CDCl_3$) of a mixture of *cis*- and *trans*-isomers of cyclobutane 3d. Green and red arrows show internuclear dipolar contacts between protons of *cis*- and *trans*-isomers correspondingly.

action with alkyl azides, such as benzyl (N_3CH_2Ph) and pivaloylmethyl (N_3CH_2OPiv) azides (Scheme 3).

The resulted 1,2,3-triazoloallenes **4a-f** were obtained in good to excellent yields (for details see Experimental section) and further tested in dimerization under found conditions (Scheme 4).

3. Conclusion

We have developed a direct catalytic access to 1,3-dimethylenecyclobutanes based on dimerization of readily available CF_3-substituted functional allenes under Ru-catalysis. The reactions smoothly proceed in toluene at 100 °C in the presence of 5 mol% naphthalene-containing ruthenium catalytic system $[Cp^*Ru(C_{10}H_8)]PF_6/[Et_3NBn]Cl$ exclusively affording tail-to-tail allene dimerization products in good to excellent yields. The corresponding 1,3-dimethylenecyclobutane derivatives are formed as mixtures of *cis,trans*-isomers that can be separated by column chromatography. Functionalized allenynes can be also applied for Ru-catalyzed dimerization after blocking of triple bond by means of triazole formation *via* Cu(I)-catalyzed azide-alkyne ligation.

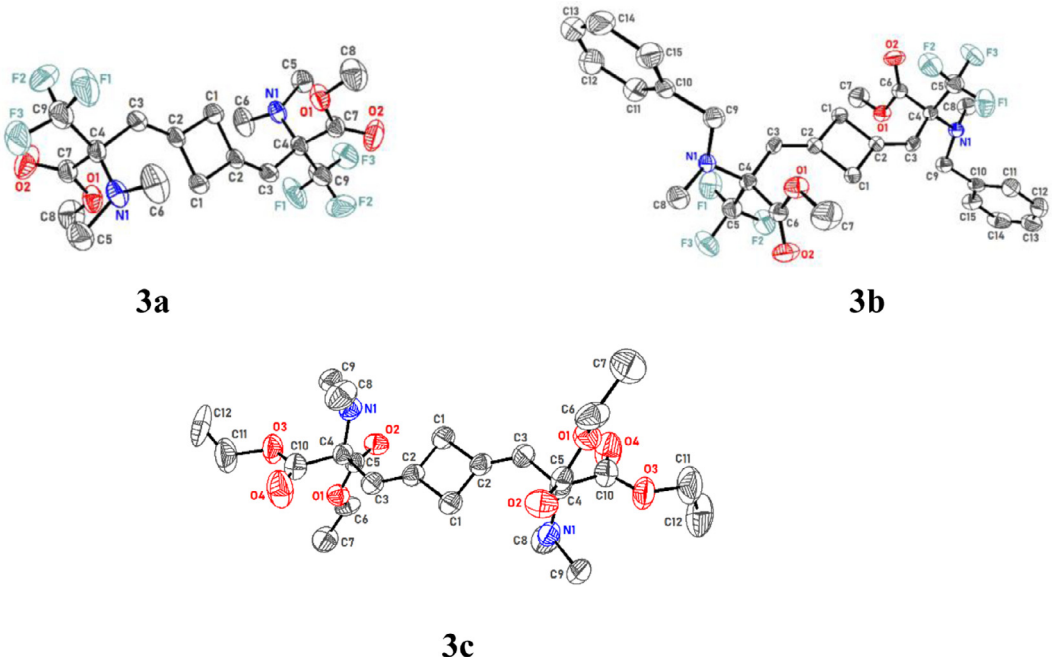
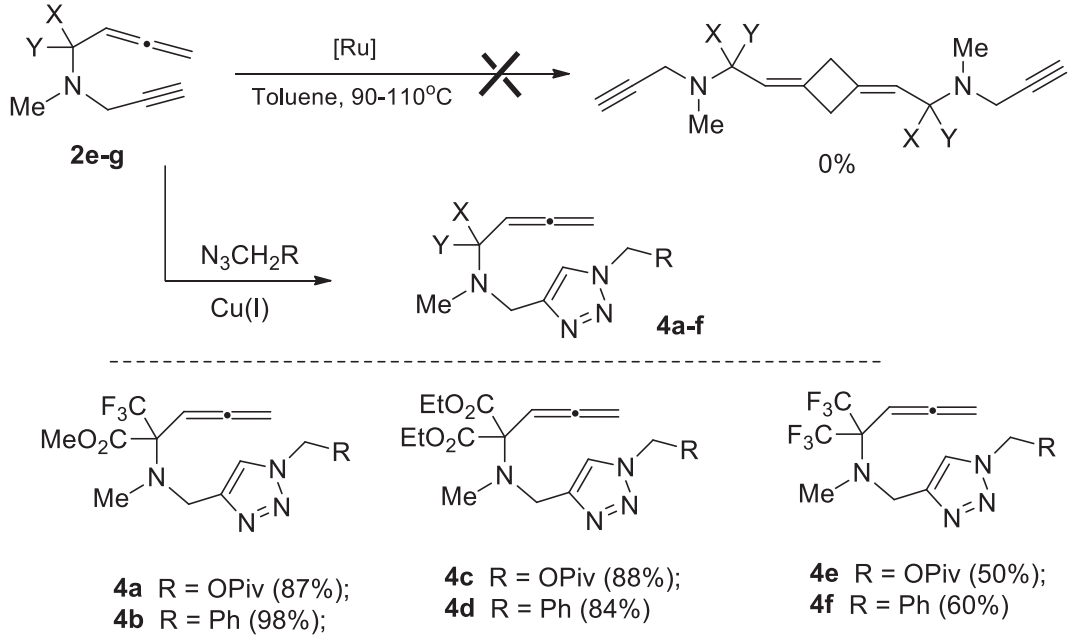


Fig. 2. Structures of cyclobutanes (*S,R*)-*E*-3a (CCDC 2081066), (*S,R*)-*E*-3b (CCDC 2081065) and 3c (CCDC 2081067). Ellipsoids are shown at the 50% level. Hydrogen atoms are omitted for clarity.



Scheme 3. Synthesis of 1,2,3-triazole-containing allenes **4a-f**.

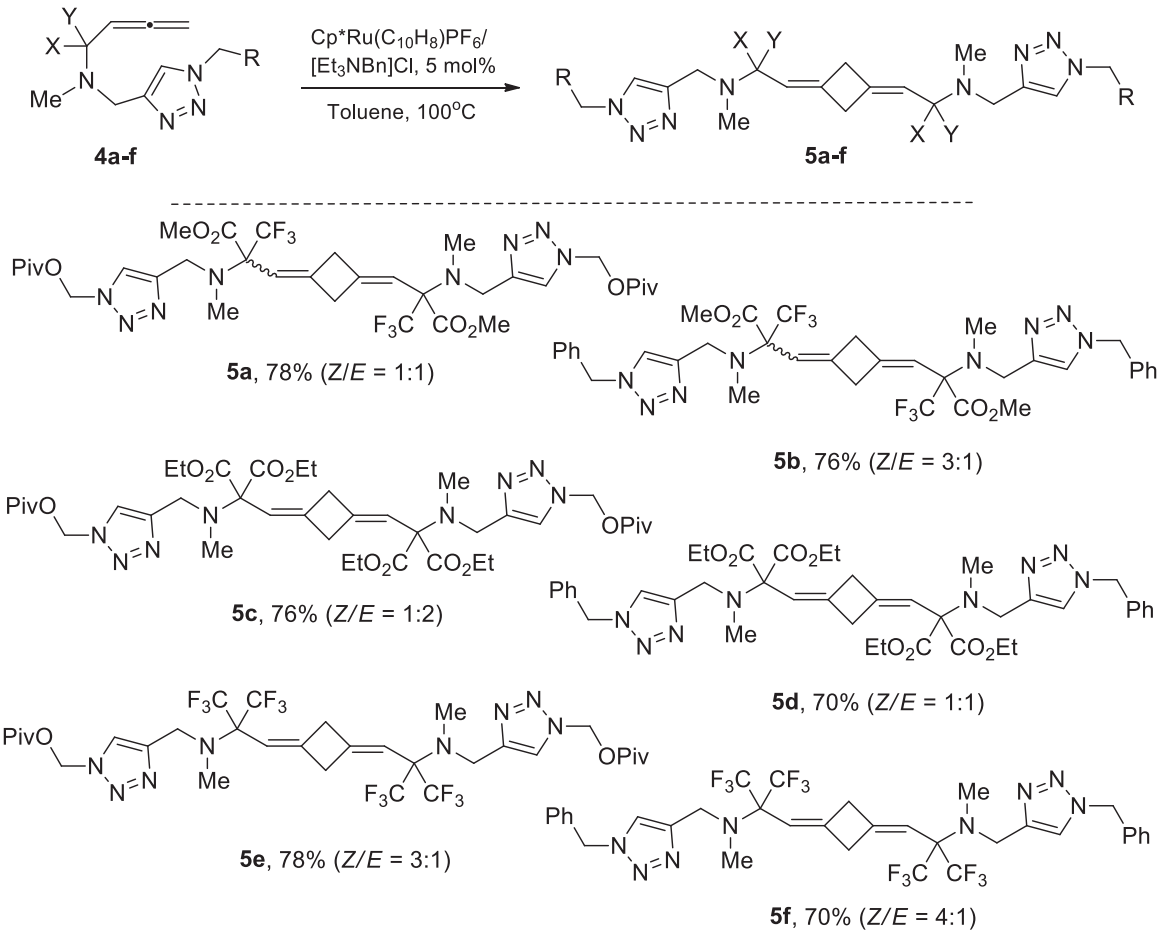
4. Experimental section

4.1. General information

All solvents used in the reactions were freshly distilled from appropriate drying agents before use. All other reagents were distilled if necessary. Analytical TLC was performed with Merck silica gel 60 F 254 plates; visualization was accomplished with UV light or spraying with $Ce(SO_4)_2$ solution in 5% H_2SO_4. Chromatography was carried out using Merck silica gel (Kieselgel 60, 0.063-0.200 mm) and petroleum ether/ethyl acetate as an eluent. The NMR spectra were obtained with Bruker AV-400, AV-500 and AV-600 spectrometers operating at 400, 500 and 600 MHz, respectively, for ^{1}H (TMS reference), at 101, 126 and 151 MHz for ^{13}C, at 376 MHz for ^{19}F (CCl_3F reference) and 161 MHz for ^{31}P (H_3PO_4 reference).

4.2. General procedure for [2,3]-sigmatropic rearrangement

A mixture of the substituted amine (210.0 mg, 11.3 mmol), copper trifluoroacetylacetonate (20.9 mg, 0.056 mmol) and the corresponding diazocompounds (94 mg, 11.3 mmol) in anhydrous toluene (5 mL) was stirred under heating (90-120°C) for 1-2 h. After the reaction completion (TLC) the solvent was removed under reduced pressure and the crude product was purified by col-



Scheme 4. Dimerization of 1,2,3-triazoloallenes **4a-f**.

umn chromatography on silica gel (eluent petroleum ether/ethyl acetate).

4.3. Characterization of the products of [2,3]-sigmatropic rearrangement

Allenes **2a** [13], **2d** [13], **2e** [13], **2f** [14c], **2g** [14c] was synthesized according to the general procedure.

Methyl 2-(benzyl(methyl)amino)-2-(trifluoromethyl)penta-3,4-dienoate (2b)

The compound was synthesized according to the general procedure. Yield 42% as a colorless oil (eluent petroleum ether/ethyl acetate = 10/1). ^{1}H NMR (400 MHz, C_6D_6) δ 7.30 (d, J = 7.5 Hz, 2H), 7.15-7.10 (m, 2H), 7.07-7.03 (m, 1H), 5.29 (t, J = 6.8 Hz, 1H), 4.53 (dd, J = 6.8 Hz, 1.1 Hz, 2H), 3.92 (d, J = 14.6 Hz, 1H), 3.84 (d, J = 14.6 Hz, 1H), 3.29 (s, 3H), 2.29 (s, 3H). ^{19}F NMR (376 MHz, C_6D_6) δ -66.79 (s, 3F). ^{13}C NMR (101 MHz, C_6D_6) δ 208.9, 167.0, 139.5, 128.2, 127.8, 126.9, 125.4 (q, J = 291.3 Hz CF_3), 88.0, 78.5, 74.4 (q, J = 24.5 Hz, >C<), 56.8, 51.6, 36.4. Anal. Calc. for $C_{15}H_{16}F_3NO_2$ (%): C, 60.20; H, 5.39; N, 4.68. Found: C, 60.21; H, 5.27; N, 4.47.

Diethyl 2-(dimethylamino)-2-(propa-1,2-dien-1-yl)malonate (2c)

The compound was synthesized according to the general procedure. Yield 75% as a yellow oil (eluent petroleum ether/ethyl acetate = 5/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.71 (t, J = 6.8 Hz, 1H), 4.97 (d, J = 6.7 Hz, 2H), 4.29 (q, J = 7.2 Hz, 4H), 2.52 (s, 6H), 1.31 (t, J = 7.1 Hz, 6H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 208.5, 167.7, 88.2, 78.2, 75.3, 61.6, 40.5, 14.1. Anal. Calc. for $C_{12}H_{19}NO_4$ (%): C, 59.73; H, 7.94; N, 5.81. Found: C, 59.51; H, 7.77; N, 5.94.

4.4. General procedure for the synthesis of triazole-containing allenes

To a solution of the corresponding acetylene-containing allene (0.14 g, 0.54 mmol) in ^{t}BuOH/H_2O (1:1 ratio) (5 mL) was added the benzyl azide or pivaloyl methyl azide (0.08g, 0.54 mmol), sodium ascorbate (21.5 mg, 0.1 mmol) and 0.5 M copper sulfate pentahydrate (0.03 mmol) sequentially. The reaction mixture was stirred at room temperature for 7 h. Upon the completion of the reaction (monitored) by TLC the mixed solvent was removed under reduced pressure and the residue was taken up in ethyl acetate (15 mL) and washed with 0.5 M HCl (2 × 10 mL). The ethyl acetate extracts were washed with water (15 mL) and dried over $MgSO_4$, filtered and the solvent removed under reduced pressure to afford crude triazole. Column chromatography (eluent petroleum ether/ethyl acetate) provided the pure target products.

4.5. Characterization of the triazole-containing allenes

Methyl 2-(methyl((1-((pivaloyloxy)methyl)-1H-1,2,3-triazol-4-yl)methyl)amino)-2-(trifluoromethyl)penta-3,4-dienoate (4a)

The compound was synthesized according to the general procedure. Yield 87% as a colorless oil (eluent petroleum ether/ethyl acetate = 4/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.78 (s, 1H), 6.24 (s, 2H), 5.41 (t, J = 6.7 Hz, 1H), 5.04 (d, J = 6.7 Hz, 2H), 4.09 (s, 2H), 3.85 (s, 3H), 2.51 (s, 3H), 1.21 (s, 9H). ^{19}F NMR (376 MHz, $CDCl_3$) δ -67.91 (s, 3F). ^{13}C NMR (101 MHz, $CDCl_3$) δ 209.2, 177.6, 167.3, 147.5, 124.7

(q, J = 290.3 Hz, CF_3), 123.6, 87.6, 79.3, 74.2 (q, J = 25.0 Hz, >C<), 69.7, 52.5, 48.7, 38.7, 37.3, 26.7. Anal. Calc. for $C_{17}H_{23}F_3N_4O_4$ (%): C, 50.49; H, 5.73; N, 13.85. Found: C, 50.39; H, 5.82; N, 13.93.

Methyl 2-(((1-benzyl-1H-1,2,3-triazol-4-yl)methyl)(methyl) amino)-2-(trifluoromethyl)penta-3,4-dienoate (4b)

The compound was synthesized according to the general procedure. Yield 98% as a colorless oil (eluent petroleum ether/ethyl acetate = 2/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.44 (s, 1H), 7.36-7.33 (m, 3H), 7.26-7.23 (m, 2H), 5.50 (s, 2H), 5.32 (t, J = 6.7 Hz, 1H), 4.94 (d, J = 6.7 Hz, 2H), 4.01 (s, 2H), 3.77 (s, 3H), 2.44 (s, 3H). ^{19}F NMR (376 MHz, $CDCl_3$) δ -67.94 (s, 3F). ^{13}C NMR (101 MHz, $CDCl_3$) δ 209.1, 167.3, 147.2, 134.6, 129.0, 128.7, 127.9, 127.5 (q, J = 288.5 Hz, CF_3), 122.4, 87.5, 79.3, 74.2 (q, J = 25.3 Hz, >C<), 54.2, 52.6, 48.8, 37.3. Anal. Calc. for $C_{18}H_{19}F_3N_4O_2$ (%): C, 56.84; H, 5.03; N, 14.73. Found: C, 56.81; H, 4.97; N, 14.66.

Diethyl 2-(methyl((1-((pivaloyloxy)methyl)-1H-1,2,3-triazol-4-yl)methyl)amino)-2-(propa-1,2-dien-1-yl)malonate (4c)

The compound was synthesized according to the general procedure. Yield 88% as a colorless oil (eluent petroleum ether/ethyl acetate = 4/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.82 (s, 1H), 6.23 (s, 2H), 5.76 (t, J = 6.7 Hz, 1H), 4.97 (d, J = 6.7 Hz, 2H), 4.29 (q, J = 7.1 Hz, 4H), 4.01 (s, 2H), 2.44 (s, 3H), 1.31 (t, J = 7.1 Hz, 6H), 1.20 (s, 9H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 208.9, 177.5, 167.9, 147.4, 124.3, 89.6, 78.5, 75.5, 69.7, 61.7, 49.0, 38.7, 37.0, 26.7, 14.1. Anal. Calc. for $C_{20}H_{30}N_4O_6$ (%): C, 56.86; H, 7.16; N, 13.26. Found: C, 56.73; H, 6.99; N, 13.03.

Diethyla0~2-(((1-benzyl-1H-1,2,3-triazol-4-yl)methyl)(methyl)amino)-2-)(propa-1,2-dien-1-ylmalonate (4d)

The compound was synthesized according to the general procedure. Yield 84% as a colorless oil (eluent petroleum ether/ethyl acetate = 2/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.83 (s, 1H), 7.38-7.35 (m, 3H), 7.30-7.28 (m, 2H), 5.83 (t, J = 6.7 Hz, 1H), 5.53 (s, 2H), 4.97 (d, J = 6.7 Hz, 2H), 4.29 (q, J = 7.1 Hz, 4H), 4.15 (s, 2H), 2.54 (s, 3H), 1.30 (t, J = 7.1 Hz, 6H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 209.2, 166.8, 149.7, 134.7, 129.1, 128.7, 128.1, 121.8, 88.7, 79.0, 75.5, 62.4, 54.2, 49.3, 37.2, 14.1. Anal. Calc. for $C_{21}H_{26}N_4O_4$ (%): C, 63.30; H, 6.58; N, 14.06. Found: C, 63.21; H, 6.77; N, 13.94.

(4-((Methyl(1,1,1-trifluoro-2-(trifluoromethyl)penta-3,4-dien-2-yl)amino)methyl)-1H-1,2,3-triazol-1-yl)methyl pivalate (4e)

The compound was synthesized according to the general procedure. Yield 50% as a colorless oil (eluent petroleum ether/ethyl acetate = 3/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.71 (s, 1H), 6.24 (s, 2H), 5.29 (t, J = 6.8 Hz, 1H), 5.09 (d, J = 6.8 Hz, 2H), 4.20 (s, 2H), 2.57 (s, 3H), 1.20 (s, 9H). ^{19}F NMR (376 MHz, $CDCl_3$) δ -66.75 (s, 6F). ^{13}C NMR (101 MHz, $CDCl_3$) δ 209.5, 177.6, 147.3, 123.9 (q, J = 294.5 Hz, CF_3), 123.3, 85.8, 79.9, 71.7-70.7 (m), 69.5, 48.0, 38.6, 36.8, 26.6. Anal. Calc. for $C_{16}H_{20}F_6N_4O_2$ (%): C, 46.38; H, 4.87; N, 13.52. Found: C, 46.32; H, 4.82; N, 13.23.

N-((1-benzyl-1H-1,2,3-triazol-4-yl)methyl)-1,1,1-trifluoro-N-methyl-2-(trifluoromethyl) penta-3,4-dien-2-amine (4f)

The compound was synthesized according to the general procedure. Yield 60% as a colorless oil (eluent petroleum ether/ethyl acetate = 2/1). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.40-7.34 (m, 3H), 7.25-7.24 (m, 3H), 5.52 (s, 2H), 5.21 (t, J = 6.8 Hz, 1H), 4.96 (d, J = 6.7 Hz, 2H), 4.14 (s, 2H), 2.53 (s, 3H). ^{19}F NMR (376 MHz, $CDCl_3$) δ -66.70 (s, 6F). ^{13}C NMR (101 MHz, $CDCl_3$) δ 209.5. 147.7, 134.7, 129.0, 128.7, 127.8, 123.9 (q, J = 293.5 Hz, CF_3), 85.8, 79.9, 71.8-70.8 (m), 54.3, 48.1, 36.9. Anal. Calc. for $C_{17}H_{16}F_6N_4$ (%): C, 52.31; H, 4.13; N, 14.35 Found: C, 52.17; H, 4.17; N, 14.56.

4.6. General procedure for allenes dimerization

A mixture of the corresponding allene (0.25 g, 11.2 mmol) and $[Cp^*Ru(C_{10}H_8)]PF_6$/$[Et_3NBn]Cl$ (28.5 mg/12.7 mg, 0.056 mmol) in anhydrous toluene (5 mL) was stirred under heating (90-120°C) for 2 h. After the reaction completion (TLC) the solvent was removed under reduced pressure and the crude product was purified by column chromatography on silica gel (eluent petroleum ether/ethyl acetate).

4.7. Characterization of 1,3methylenecyclobutanes

Dimethyl 2,2′-(cyclobutane-1,3-diylidenebis(methanylylidene)) bis(2-(dimethylamino)-3,3,3-trifluoropropanoate) (3a)

The compound was synthesized according to the general procedure. Yield 71% as a colorless oil (eluent petroleum ether/ethyl acetate = 10/1) as a mixture of diastereomers *A* and *B* at the ratio 1:1. ***Diastereomer A***, yield 30%. Mixture of *Z*/*E* isomers in ratio of 1:1. ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.70 (s, 1H).5.40-5.39 (m, 1H), 3.83 and 3.82 (both s, 6H), 3.47-3.36 (m, 2H), 2.71-2.64 (m, 2H), 2.50 and 2.49 (both s, 12H). ^{19}F NMR (376 MHz, $CDCl_3$) δ: -67.42 and -67.02 (both s, 6F). ^{13}C NMR (151 MHz, $CDCl_3$) δ 167.8, 167.6, 148.1, 143.7, 125.3 (q, J = 289.5 Hz, CF_3), 115.3, 112.3, 73.3-72.8 (m, >C<), 52.5, 52.3, 40.5, 40.3, 28.0. ***Diastereomer B***, yield 30%. Mixture of *E*/*Z* isomers in ratio of 1:1. ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.71 (s, 1H).5.40-5.39 (m, 1H), 3.83 and 3.82 (both s, 6H), 3.42 (s, 2H), 2.71-2.65 (m, 2H), 2.50 and 2.49 (both s, 12H). ^{19}F NMR (376 MHz, $CDCl_3$) δ -67.33 and -66.92 (both s, 6F). ^{13}C NMR (151 MHz, $CDCl_3$) δ 167.8, 167.6, 148.1, 143.6, 125.4 (q, J = 290.4 Hz, CF_3), 125.3 (q, J = 290.1 Hz, CF_3), 115.3, 112.3, 73.2-72.6 (m, >C<), 52.4, 52.3, 40.4, 40.3, 27.9. Anal. Calc. for $C_{18}H_{24}F_6N_2O_4$ (%): C, 48.43; H, 5.42; N, 6.28. Found: C, 48.21; H, 5.27; N, 6.14.

Dimethyl 2,2′-(cyclobutane-1,3-diylidenebis(methanylylidene)) bis(2-(benzyl(methyl)amino)-3,3,3-trifluoropropanoate) (3b)

The compound was synthesized according to the general procedure. Yield 85% as a colorless oil (eluent petroleum ether/ethyl acetate = 16/1) as a mixture of diastereomers at the ratio 1:1. ***E-isomer***, yield 12%. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.41-7.39 (m, 4H), 7.37-7.33 (m, 4H), 7.30-7.26 (m, 2H), 5.51-5.50 (m, 2H),3.97 (dd, J = 14.3 Hz, 3.4 Hz, 2H), 3.86 and 3.85 (both s, 6H), 3.77 (d, J = 14.3 Hz, 2H), 3.59-3.51 (m, 2H), 3.47-3.39 (m, 2H), 2.36 and 2.35 (both s, 6H). ^{19}F NMR (376 MHz, $CDCl_3$) δ -67.40 and -67.35 (both s, 6F). ^{13}C NMR (126 MHz, $CDCl_3$) δ 168.0, 144.1, 139.4, 128.4, 128.1, 127.1, 125.5 (q, J = 292.5 Hz, CF_3), 115.5, 73.8 (q, J = 24.8 Hz, >C<), 57.3, 52.6, 40.7, 36.4. ***Z-isomer***, yield 25%. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.39-7.36 (m, 4H), 7.33-7.29 (m, 4H), 7.26-7.22 (m, 2H), 5.73 (s, 2H), 3.92 (d, J = 14.0 Hz, 2H), 3.84 and 3.83 (both s, 6H), 3.74 (d, J = 14.5 Hz, 2H), 2.84-2.74 (m, 2H), 2.73-2.64 (m, 2H), 2.33 and 2.32 (both s, 6H). ^{19}F NMR (376 MHz, $CDCl_3$) δ -67.07 and -66.99 (both s, 6F). ^{13}C NMR (126 MHz, $CDCl_3$) δ 167.7, 148.5, 148.4, 139.3, 128.3, 128.0, 127.0, 125.4 (q, J = 293.5 Hz, CF_3), 112.4, 112.3, 73.6 (q, J = 24.9 Hz, >C<), 57.4, 52.6, 36.5, 28.3, 28.2. Anal. Calc. for $C_{30}H_{32}F_6N_2O_4$ (%): C, 60.20; H, 5.39; N, 4.68. Found: C, 60.21; H, 5.47; N, 4.84.

Tetraethyl 2,2′-(cyclobutane-1,3-diylidenebis(methanylylidene)) bis(2-(dimethylamino) malonate) (3c)

The compound was synthesized according to the general procedure. Yield 55% as a pale yellow oil (eluent petroleum ether/ethyl acetate = 4/1). Mixture of *Z*/*E* isomers in ratio of 2:3. ^{1}H NMR (400 MHz, $CDCl_3$) δ 6.05 (s, 1H).5.70 (s, 1H), 4.33-4.21 (m, 8H), 3.44 (s, 2H), 2.67 (s, 2H), 2.46 (br. s, 12H), 1.31 (t, J = 7.1 Hz, 12H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 167.6, 147.4, 142.9, 116.7, 114.0, 74.9, 74.8, 61.5, 61.4, 40.6, 40.5, 40.4, 28.0, 14.1. Anal. Calc. for $C_{24}H_{38}N_2O_8$ (%): C, 59.73; H, 7.94; N, 5.81. Found: C, 59.69; H, 7.73; N, 5.75.

Tetraethyl cyclobutane-1,3-diylidenebis(2-(dimethylamino)-1,1,1-trifluoropropan-2-yl-3-ylidene)bis(phosphonate) (3d)

The compound was synthesized according to the general procedure. Yield 86% as a pale yellow oil (eluent petroleum ether/ethyl acetate = 2/1) as a mixture of diastereomers *A* and *B* at the ratio 1:1. ***Diastereomer A***, yield 40%. Mixture of *Z*/*E* isomers in ratio of 1:1. ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.55 (s, 1H), 5.25-5.22 (m, 1H), 4.31-4.10 (m, 8H), 3.89-3.77 (m, 1H), 3.55-3.43 (m, 1H), 3.12-

3.05 (m, 1H), 2.86-2.77 (m, 1H), 2.65 (s, 12H), 1.42-1.34 (m, 12H). ^{19}F NMR (376 MHz, $CDCl_3$) δ -63.93 and -63.65 (both s, 6F). ^{31}P NMR (161 MHz, $CDCl_3$) δ 16.72 and 16.52. ^{13}C NMR (151 MHz, $CDCl_3$) δ 149.0, 145.2, 125.9 (q, J = 297.3 Hz, CF_3), 112.5, 110.4, 71.8-71.3 (m, >C<), 70.1- 69.7 (m, >C<), 64.3, 62.8, 41.7, 40.9, 29.2, 16.5-16.1 (m). ***Diastereomer B, Z-isomer***, yield 21%. ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.57 (s, 2H), 4.34-4.09 (m, 8H), 3.11-3.01 (m, 2H), 2.83-2.73 (m, 2H), 2.66 (s, 12H), 1.42-1.36 (m, 12H). ^{19}F NMR (376 MHz, $CDCl_3$) δ -63.22 (s, 6F). ^{31}P NMR (161 MHz, $CDCl_3$) δ 16.76. ^{13}C NMR (151 MHz, $CDCl_3$) δ 149.0, 125.9 (q, J = 294.7 Hz, CF_3), 110.5, 71.7 (q, J = 25.4 Hz, >C<,), 64.1, 63.1, 41.2, 29.6, 29.3, 16.5, 16.3. Anal. Calc. for $C_{22}H_{38}F_6N_2O_6P_2$ (%): C, 43.86; H, 6.36; N, 4.65. Found: C, 43.29; H, 6.13; N, 4.61.

Dimethyl 2,2′-(cyclobutane-1,3-diylidenebis(methanylylidene)) bis(3,3,3-trifluoro-2-(methyl((1-((pivaloyloxy)methyl)-1H-1,2,3-triazol-4-yl)methyl)amino)propanoate) (5a)

The compound was synthesized according to the general procedure. Yield 78% as a colorless oil (eluent petroleum ether/ethyl acetate = 8/1) as a mixture of diastereomers *A* and *B* at the ratio 1:1. Mixture of *Z/E* isomers in ratio of ~1:1. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.81 (s, 2H), 6.25 (s, 4H), 5.76 (s, 1H), 5.49 (s, 1H), 4.08-3.99 (m, 4H), 3.85 (s, 6H), 3.52-3.35 (m, 2H), 2.75-2.63 (m, 2H), 2.45 (s, 6H), 1.20 (s, 18H). ^{19}F NMR (376 MHz, $CDCl_3$) δ -67.79 and -67.76 (both s, 6F), -67.45 and -67.40 (both s, 6F). ^{13}C NMR (101 MHz, $CDCl_3$) δ 177.6, 167.7, 167.5, 148.5, 147.5, 147.4, 144.4, 144.3, 125.3 (q, J = 290.1 Hz, CF_3), 125.2 (q, J = 288.8 Hz, CF_3), 123.8, 115.1, 115.0, 112.2, 73.4, (q, J = 25.1 Hz, >C<), 73.2, (q, J = 25.0 Hz, >C<), 52.7, 52.6, 48.8, 48.7, 40.5, 38.7, 36.9, 36.7, 27.9, 26.7. Anal. Calc. for $C_{34}H_{46}F_6N_8O_8$ (%): C, 50.49; H, 5.73; N, 13.85. Found: C, 50.22; H, 5.77; N, 14.04.

Dimethyl 2,2′-(cyclobutane-1,3-diylidenebis(methanylylidene)) bis(2-(((1-benzyl-1H-1,2,3-triazol-4-yl)methyl)(methyl)amino)-3,3,3-trifluoropropanoate) (5b)

The compound was synthesized according to the general procedure. Yield 73% as a pale orange oil (eluent petroleum ether/ethyl acetate = 5/1), as a mixture of diastereomers *A* and *B* at the ratio 1:1. Mixture of *Z/E* isomers in ratio of ~1:1. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.46-7.43 (m, 2H), 7.46-7.32 (m, 6H), 7.25-7.23 (m, 4H), 5.66 (s, 1H), 5.50 (s, 4H), 5.33 (s, 1H), 3.98-3.88 (m, 4H), 3.77 and 3.75 (both s, 6H), 3.35-3.17 (m, 2H), 2.63-2.47 (m, 2H), 2.37 (s, 6H). ^{19}F NMR (376 MHz, $CDCl_3$) δ -67.94 (s, 6F), -67.51 (s, 6F). ^{13}C NMR (101 MHz, $CDCl_3$) δ 167.6, 148.3, 147.0, 144.1, 134.6, 129.0, 128.6, 127.9, 125.2 (q, J = 275 Hz, CF_3), 122.6-122.4 (m), 115.1, 112.3, 73.1-72.9 (m), 54.2, 52.7, 52.6, 48.9, 48.8, 40.3, 37.0, 36.8, 27.7. Anal. Calc. for $C_{36}H_{38}F_6N_8O_4$ (%): C, 56.84; H, 5.03; N, 14.73 Found: C, 56.71; H, 4.86; N, 14.54.

Tetraethyl 2,2′-(cyclobutane-1,3-diylidenebis(methanylylidene)) bis(2-(methyl((1-((pivaloyloxy)methyl)-1H-1,2,3-triazol-4-yl)methyl)amino)malonate) (5c)

The compound was synthesized according to the general procedure. Yield 76% as a pale orange oil (eluent petroleum ether/ethyl acetate = 8/1). Mixture of *Z/E* isomers in ratio of ~1:2. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.81 and 7.79 (both s, 2H), 6.17 and 6.16 (both s, 4H), 6.04 (s, 1H), 5.73 (s, 1H), 4.25-4.11 (m, 8H), 3.83 (s, 4H), 3.35 (s, 2H), 2.61 (s, 2H), 2.17 (s, 6H), 1.23-1.17 (m, 12H), 1.11 and 1.10 (both s, 18H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 177.4, 167.9, 167.8, 147.7, 143.7, 124.2, 124.1, 116.9, 114.6, 75.4, 75.3, 69.7, 69.6, 61.6, 61.5, 49.5, 49.4, 40.7, 38.6, 36.5, 36.3, 28.5, 26.7, 14.1. Anal. Calc. for $C_{40}H_{60}N_8O_{12}$ (%): C, 56.86; H, 7.16; N, 13.26. Found: C, 56.81; H, 6.97; N, 13.10.

Tetraethyl 2,2′-(cyclobutane-1,3-diylidenebis(methanylylidene)) bis(2-(((1-benzyl-1H-1,2,3-triazol-4-yl)methyl)(methyl)amino) malonate) (5d)

The compound was synthesized according to the general procedure. Yield 70% as a pale yellow oil (eluent petroleum ether/ethyl acetate = 6/1). Mixture of *Z/E* isomers in ratio of ~1:1. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.53 (s, 2H), 7.36-7.20 (m, 10H), 6.02 (s, 1H), 5.68 (s, 1H), 5.48 (s, 4H), 4.24-4.12 (m, 8H), 3.83 (s, 4H), 3.22 (s, 2H), 2.51 (s, 2H), 2.19 (s, 6H), 1.21 (td, J = 7.1 Hz, 2.0 Hz, 12H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 167.8, 147.7, 143.5, 141.04, 134.8, 134.7, 129.0, 128.9, 128.6, 128.5, 127.9, 127.8, 122.9, 116.8, 114.6, 75.3, 61.6, 54.1 and 54.0, 49.6 and 49.5, 40.7, 36.5, 36.3, 28.4, 14.1. Anal. Calc. for $C_{42}H_{52}N_8O_8$ (%): C, 63.30; H, 6.58; N, 14.06. Found: C, 63.41; H, 6.67; N, 13.98.

(4,4′-((((Cyclobutane-1,3-diylidenebis(methanylylidene)) bis(1,1,1,3,3,3-hexafluoropropane-2,2-diyl))bis(methylazanediyl)) bis(methylene))bis(1H-1,2,3-triazole-4,1-diyl))bis(methylene) bis(2,2-dimethylpropanoate) (5e)

The compound was synthesized according to the general procedure. Yield 78% as a colorless oil (eluent petroleum ether/ethyl acetate = 3/1). Mixture of *Z/E* isomers in ratio of 3:1. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.70 and 7.71 (both s, 2H), 6.26 (s, 4H), 5.66 (s, 1H), 5.36 (s, 1H), 4.14 (s, 4H), 3.54 (s, 2H), 2.81 (s, 2H), 2.51 and 2.48 (both s, 6H), 1.20 (s, 18H). ^{19}F NMR (376 MHz, $CDCl_3$) δ -66.30 (s, 12F) and -65.81 (s, 12F). ^{13}C NMR (101 MHz, $CDCl_3$) δ 177.8, 177.7, 149.8, 146.4, 146.2, 146.1, 123.7, 123.6, 124.5 (q, J = 295.2 Hz, CF_3), 124.4 (q, J = 295.2 Hz, CF_3), 112.4, 110.1, 71.9-70.8 (m), 69.6, 47.3, 41.5, 38.7, 36.9, 36.8, 29.1, 26.6. Anal. Calc. for $C_{32}H_{40}F_{12}N_8O_4$ (%): C, 46.38; H, 4.87; N, 13.52. Found: C, 46.28; H, 4.77; N, 13.44.

2,2′-(Cyclobutane-1,3-diylidenebis(methanylylidene))bis(N-((1-benzyl-1H-1,2,3-triazol-4-yl)methyl)-1,1,1,3,3,3-hexafluoro-N-methylpropan-2-amine) (5f)

The compound was synthesized according to the general procedure. Yield 70% as a pale yellow oil (eluent petroleum ether/ethyl acetate = 3/1). Mixture of *Z/E* isomers in ratio of 4:1. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.40-7.37 (m, 8H), 7.30-7.26 (m, 4H), 5.61 (s, 1H), 5.56 (s, 4H), 5.18 (s, 1H), 4.09 (s, 4H), 3.34 (s, 2H), 2.71 (s, 2H), 2.48 and 2.45 (both s, 6H). ^{19}F NMR (376 MHz, $CDCl_3$) δ -66.30 (s, 12F), -65.80 (s, 12F). ^{13}C NMR (101 MHz, $CDCl_3$) δ 149.8, 146.2, 146.0, 134.7, 129.1, 128.8, 128.6, 127.9, 127.8, 124.5 (q, J = 294.9 Hz, CF_3), 124.4 (q, J = 294.5 Hz, CF_3), 122.2, 122.0, 112.4, 110.1, 71.9-70.8 (m), 54.1, 47.3, 41.3, 36.8, 36.7, 29.1. Anal. Calc. for $C_{34}H_{32}F_{12}N_8$ (%): C, 52.31; H, 4.13; N, 14.35. Found: C, 52.19; H, 4.06; N, 14.51.

4.8. X-ray structure analysis of 3a-3c: Bruker AXS SMART 1000, CCD-detector λ(MoKa) = 0.71073Å, graphite monochromator, ω-scanning, T=150

The structure was solved by direct method and refined by full-matrix least squares method for F^2 with anisotropic parameters for all non-hydrogen atoms. All calculations were performed with the use of the SHELXTL PLUS program packages [18–21]. XRD experiment and refinement are detailed in Supporting Information (S58-S62). Data CCDC 2081066 (**3a**), 2081065 (**3b**), 2081067 (**3c**) contains the supplementary crystallographic information for this paper. These data can be obtained free of charge from the Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jorganchem.2021.121998.

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